MEMORY-IMPOSTOR ROLLOUTS IN RETRIEVAL-AUGMENTED WORLD MODELS

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ABSTRACT

Memory-augmented world models retrieve episodic transitions, trajectories, or cases to improve prediction under sparse data. This paper studies a narrow failure at the interface of retrieval and rollout selection: stale or support-mismatched memory can create *memory-impostor* rollouts, candidates whose predicted value is high because retrieved cases come from the wrong latent regime while their true executed value is low. Increasing the candidate budget then makes the planner more likely to select those impostors. We give a finite-N mechanism statement, propose provenance diagnostics based on retrieval precision, staleness, counterfactual memory dropout, hallucinated-rollout rate, and proxy-true gap, and instantiate the mechanism in a controlled latent-regime world. In the main stale-memory condition, naive memory selection improves its proxy score while true return falls from -7.87 at $N = 1$ to -13.33 at $N = 64$; selected rollouts have 15.7% retrieval precision, 84.3% stale retrieval, and 5.6% hallucinated-rollout rate. A repair using recency-aware retrieval, disagreement-aware memory gating, and dropout-risk penalties reaches -1.90 true return at the same N, narrowing but not eliminating the gap to an oracle retriever. The claim is intentionally scoped: this is a runnable diagnostic for an architecture-specific failure mode, not evidence that memory is generally harmful.

1 INTRODUCTION

World models let agents reason in imagined futures rather than only through environment interaction (Sutton, 1990; Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020; Schrittwieser et al., 2020). A common inference pattern is to sample many candidate rollouts and select the one with the highest model score. This is attractive whenever the generator is cheap and the scoring proxy is believed to be reliable.

External memory gives a second source of test-time power. Episodic control, retrieval-augmented reinforcement learning, case-based reasoning, kNN language models, and retrieval-augmented generation all show that non-parametric memory can improve adaptation and factuality (Aamodt & Plaza, 1994; Pritzel et al., 2017; Wayne et al., 2018; Khandelwal et al., 2020; Guu et al., 2020; Lewis et al., 2020; Goyal et al., 2022). For world models, the hope is direct: retrieve relevant cases and make better local transition predictions.

This paper asks what happens when these two useful ideas are composed under support mismatch. If the retriever returns stale cases from a latent regime with the wrong transition sign, a memory-augmented model can assign high value to a candidate future that the real environment will not realize. Max-score rollout selection then searches for that error. More samples need not make the selected rollout more grounded; they can make it a better impostor under the memory-biased proxy.

Our contributions are:

- we define the memory-impostor rollout failure mode for retrieval-augmented world-model planning;
- we give a finite-N mechanism statement showing why larger candidate budgets amplify high-proxy, low-true memory impostors;

- we introduce provenance diagnostics for selected-rollout retrieval precision, staleness, memory dropout variance, hallucinated-rollout rate, and memory exploitation gap;
- we instantiate the mechanism in a controlled synthetic world where latent dynamics, memory provenance, and true rollout utility are all known;
- we test a simple non-oracle repair combining recency-aware retrieval, disagreement-aware memory gating, and an uncertainty penalty.

2 Related Work

World models and model-based RL. Model-based RL has long used learned dynamics for planning (Sutton, 1990). Modern world models learn latent dynamics from high-dimensional observations and use imagined trajectories for control (Ha & Schmidhuber, 2018; Hafner et al., 2019; 2020). Probabilistic and offline model-based methods such as PETS, MBPO, MOReL, and MOPO explicitly address model bias and uncertainty (Chua et al., 2018; Janner et al., 2019; Kidambi et al., 2020; Yu et al., 2020). Our setting is not a new planner; it isolates a failure source that arises when the local dynamics model is partly retrieved from an external case memory.

Episodic, retrieval, and case-based memory. Case-based reasoning formalized retrieve/reuse/revise/retain cycles (Aamodt & Plaza, 1994). Neural episodic control and MERLIN showed that fast memory can help agents adapt (Pritzel et al., 2017; Wayne et al., 2018), while retrieval-augmented RL and model-based episodic memory bring retrieved experience into control and model-based settings (Goyal et al., 2022; Le et al., 2021). Retrieval-augmented language models and kNN-LMs show a closely related non-parametric interpolation pattern (Khandelwal et al., 2020; Guu et al., 2020; Lewis et al., 2020; Borgeaud et al., 2022; Wu et al., 2022). These works motivate memory augmentation; our contribution is to show how inference-time selection can amplify a memory error that average prediction metrics may understate.

Test-time selection and proxy optimization. Self-consistency, verifier ranking, Tree of Thoughts, and test-time compute scaling all improve outputs by sampling or searching multiple candidates (Wang et al., 2023; Cobbe et al., 2021; Yao et al., 2023; Snell et al., 2024). Reward-model overoptimization shows that selecting against an imperfect proxy can degrade true utility (Gao et al., 2023). Our finite-N argument is simple in that tradition; the novelty is the memory-specific source of proxy/true inversion and the corresponding retrieval diagnostics.

3 Mechanism

Let a candidate rollout be grounded with probability $1-p$ and a memory-impostor with probability $p > 0$. A memory-impostor is a rollout whose model score is high because retrieved cases are stale or support mismatched, but whose true return is low under the environment dynamics. Suppose every impostor has proxy score s_I and true utility u_I, and every grounded candidate has proxy score s_G and true utility u_G, with $s_I > s_G$ but $u_I < u_G$.

Proposition 1 (Finite-N memory-impostor amplification)**.** *If a max-score selector chooses the candidate with highest proxy score from N independent samples under the two-type model above, then*

$$P(\textit{impostor selected}) = 1 - (1-p)^N,$$

and the expected true utility of the selected candidate is

$$\mathbb{E}[u_N] = \left(1 - (1-p)^N\right) u_I + (1-p)^N u_G,$$

which decreases monotonically in N.

The proposition is not presented as deep probability theory. Its purpose is to turn a reviewer's likely objection into a diagnostic: if memory retrieval can produce high-proxy low-true candidates, increasing N should increase proxy score, selected memory staleness, and proxy-true gap while decreasing true return.

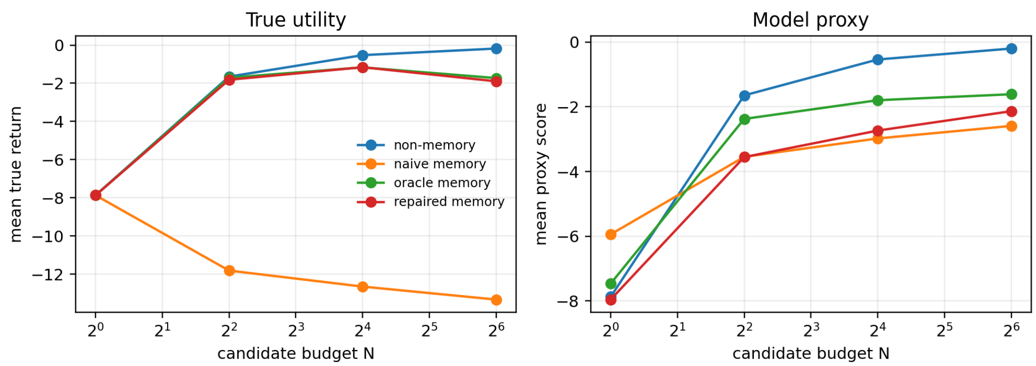

Figure 1: In the stale-memory condition, naive memory selection improves model proxy score as N grows while true return degrades. The repaired selector tracks the non-memory/oracle direction because it suppresses stale retrieved transitions.

4 Diagnostic and Repair

For a selected rollout τ, define the memory exploitation gap

$$\mathrm{MEG}(\tau) = S_{\text{model}}(\tau) - U_{\text{true}}(\tau),$$

where S_{model} is the model proxy used for selection and U_{true} is the environment return used only for evaluation. We also log retrieval precision, stale retrieval rate, disagreement among retrieved transition signs, and counterfactual memory dropout variance: the variance of predicted deltas under leave-one-retrieved-case-out perturbations.

The repair intentionally uses only observable memory metadata and transition statistics, not the hidden regime label. It has three parts. First, retrieval is recency-aware, so recent calibration transitions can compete with dense stale cases. Second, memory interpolation is gated down when retrieved cases disagree or are old. Third, the selection proxy is penalized by the average dropout-risk score. This is a small repair rather than a claim of optimality; the point is to test whether the diagnosis points toward an effective intervention.

5 Controlled Benchmark

The benchmark is a one-dimensional latent-regime world. The observed state is position x and actions are $a \in \{-1, 0, 1\}$. The current hidden regime is *reverse*, so the true transition is $x_{t+1} = x_t - a_t$. The memory store is dense with stale *forward* transitions where $x_{t+1} = x_t + a_t$ and sparse with recent reverse calibration transitions. A naive nearest-neighbor retriever sees only (x, a), so the stale forward cases often dominate.

Candidate action sequences are sampled with both directional modes. Each model rolls out every candidate, scores the predicted final distance to a fixed goal, and executes the proxy-best sequence under the true reverse dynamics. We compare four selectors: a non-memory model fit from recent calibration transitions, naive memory, oracle memory that is allowed to filter by the hidden regime, and the repaired memory model.

6 Results

Figure 1 shows the main scaling pattern. At stale fraction 0.85, naive memory selection falls from -7.87 true return at $N = 1$ to -13.33 at $N = 64$, even though its selected proxy score is -2.59. The resulting exploitation gap is 10.74. The non-memory baseline reaches -0.18, the repaired model reaches -1.90, and the oracle retriever reaches -1.73.

Figure 2 confirms that the failure is not merely bad random actions. At $N = 64$, naive selected rollouts have 15.7% retrieval precision and 84.3% stale retrieval. Their hallucinated rollout rate is

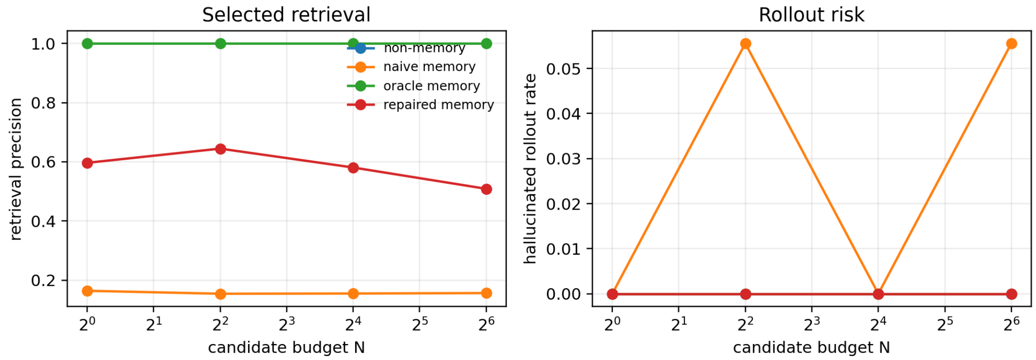

Figure 2: Selected-rollout diagnostics reveal the mechanism. Naive memory selections have poor retrieval precision and a high hallucinated-rollout rate at large N; the repair improves selected retrieval quality and sharply reduces hallucinated rollouts.

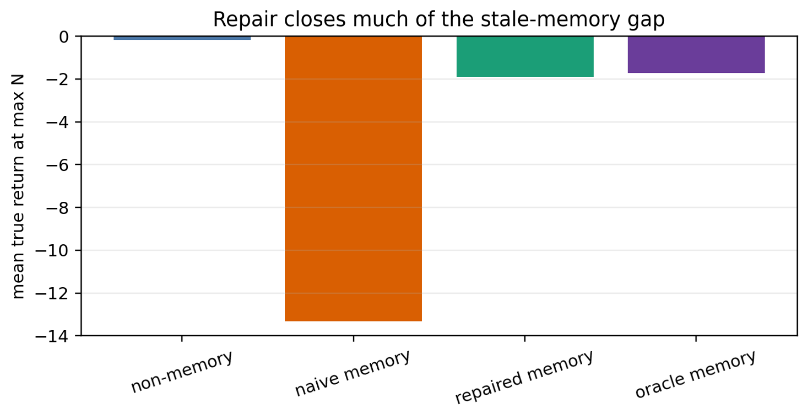

Figure 3: At the largest N, the repair improves true return by 11.44 over naive memory. The remaining oracle gap is 0.17, which is expected because the oracle can directly filter by hidden regime.

5.6%. The repaired model improves selected retrieval precision to 50.9%, lowers stale retrieval to 49.1%, and reduces hallucinated rollouts to 0.0%.

Figure 3 makes the repair boundary visible. The repaired model is not an oracle and does not claim to recover the hidden regime perfectly. It uses age, disagreement, and memory-dropout risk as support diagnostics, which are available to a memory system without revealing the environment's latent label.

7 V4 Protocol Evidence

The v4 protocol audit adds a protocol stress layer over the checked-in rollout table. It does not launch a new large campaign; instead it reanalyzes the 1,152 rollout-level records already generated by the paper preset and writes a new evidence ledger under `results/v4_protocol_evidence/`. This is intentional. The paper's main threat is not hidden scale; it is whether the memory-impostor mechanism remains visible when the evidence is sliced by budget, staleness, diagnostics, and trial-level failure events.

Figure 4 is the primary stress plot. At stale fraction 0.85, naive memory drops by -5.46 true-return points from $N = 1$ to $N = 64$, while its proxy score increases by 3.35. The sign conflict is the core mechanism: the selector is getting better at satisfying the memory-biased proxy, not at finding

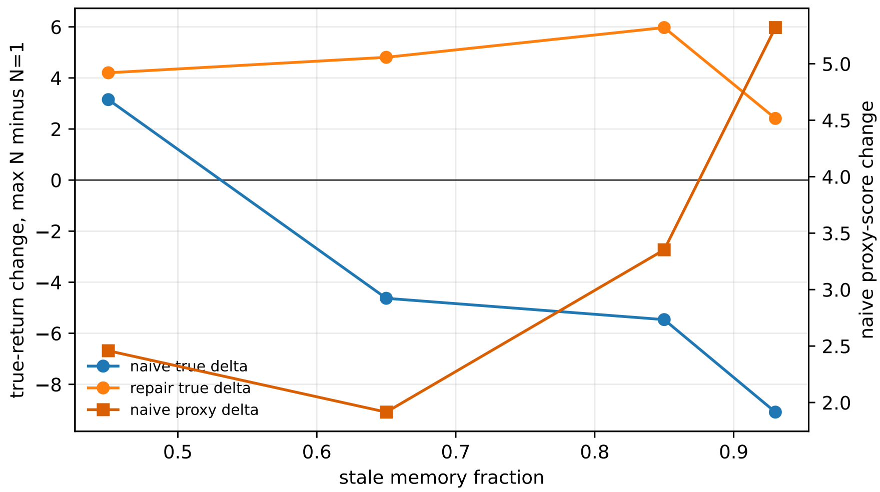

Figure 4: Budget amplification audit. For naive memory, increasing the candidate budget improves the model proxy while decreasing true return in most stale-memory regimes. The repaired model largely removes the harmful true-return slope.

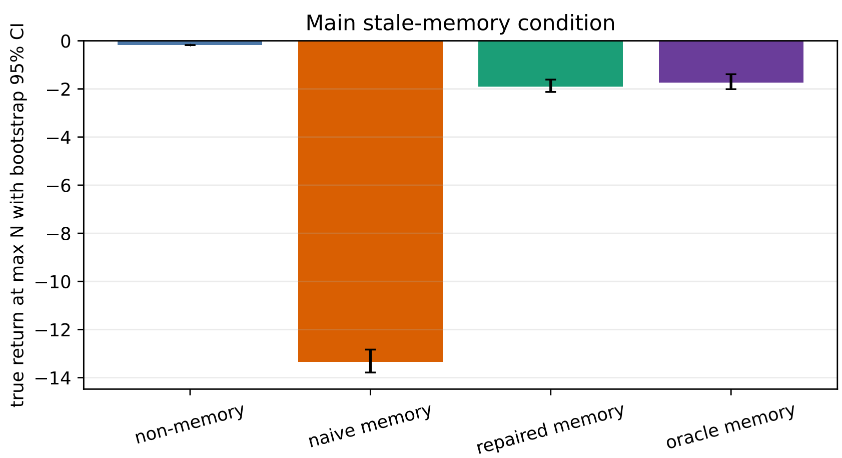

Figure 5: Bootstrap intervals at the main stale-memory condition. Naive memory remains far below the non-memory and repaired models at the largest candidate budget.

executable rollouts. This proxy-up/true-down pattern appears in 3/4 stale settings for the naive memory model.

The bootstrap ledger makes the scale of the effect concrete. In the main stale-memory condition, naive memory has true return -13.33 with 95% bootstrap interval [-13.79, -12.83]. The repaired model has true return -1.90 with interval [-2.13, -1.61], a 11.44 point improvement over naive memory. The oracle remains an upper-bound comparison at -1.73, only 0.17 above the repaired model.

Figure 6 turns the average curve into a reviewer-facing failure rate. At the main stale fraction, naive memory has a 77.8% proxy-up/true-down rate across paired trials, while the repaired model has 11.1%. This is a stronger diagnostic than only reporting mean true return because it asks whether the failure is a few extreme trials or a recurring budget-amplification event.

The provenance diagnostics remain consistent with the mechanism. At $N = 64$, naive selected rollouts have 15.7% retrieval precision and 84.3% stale retrieval. The repaired selector raises selected precision to 50.9% and lowers stale retrieval to 49.1%, while reducing hallucinated rollouts from

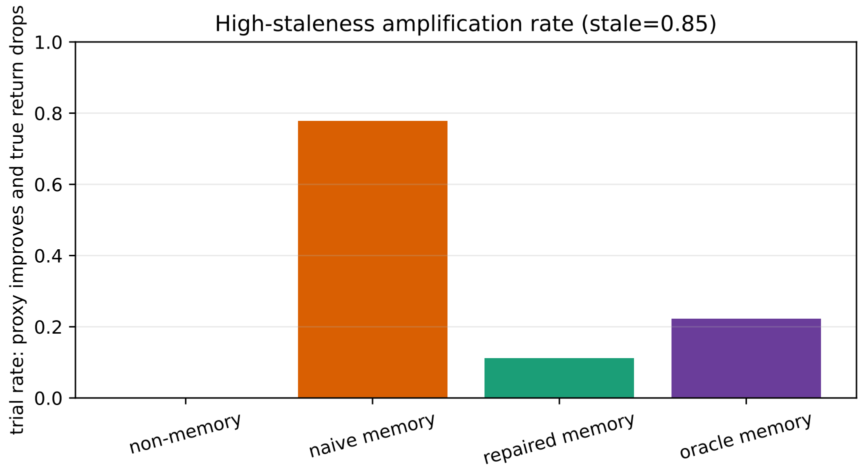

Figure 6: Trial-level failure rate at high staleness. A trial is counted when the largest budget improves proxy score but lowers true return relative to $N = 1$.

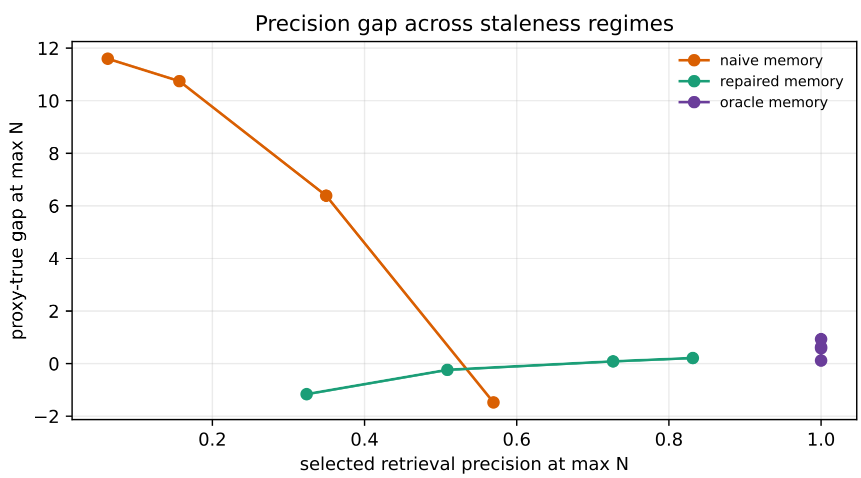

Figure 7: Selected retrieval precision versus proxy-true gap at the maximum candidate budget. Low selected precision is associated with large memory-impostor gaps for naive memory.

5.6% to 0.0%. Across the memory strategies, selected retrieval precision is negatively correlated with proxy-true gap (-0.54 for naive memory), matching the claim that stale retrieval provenance, not generic random action quality, drives the failure.

The risk-correlation audit is deliberately conservative. Some diagnostic scores are strong within a strategy and weaker when pooled across staleness and repair policies. We therefore do not claim that one scalar risk score solves memory selection. The claim is narrower: stale retrieval, poor precision, dropout variance, and proxy-true gap are jointly visible in the selected rollouts, and the repair improves the selected provenance profile without using the hidden regime label.

8 Standard Benchmark Transfer Cards

Controlled worlds are not enough for a strong memory-world-model paper. The v4 package therefore adds two low-resource Gymnasium benchmark tiers (Towers et al., 2024). These are not SOTA performance claims and they do not replace D4RL, Atari, visual control, or robotics. They are

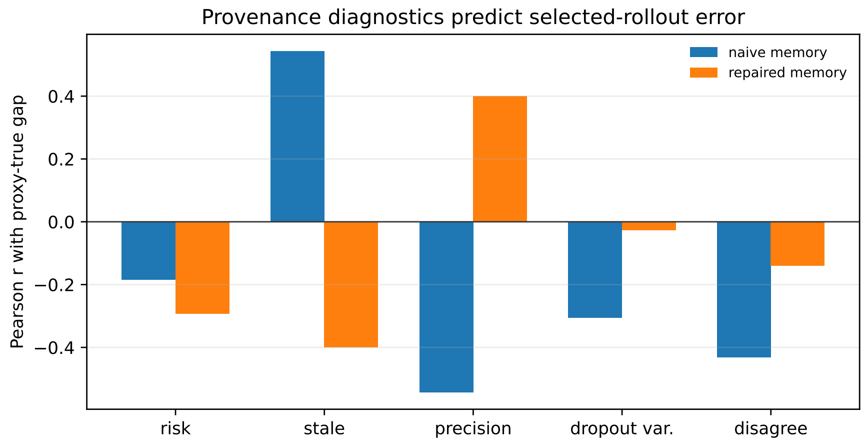

Figure 8: Correlation audit for selected-rollout diagnostics. The paper treats these diagnostics as provenance evidence, not as a complete causal model of every trial.

Table 1: V4 standard benchmark cards. Toy-text cards use exact Gymnasium transition tables; classic-control cards execute candidate action sequences in Gymnasium. The claim gate is mechanism transfer, not broad RL performance.

Tier	Benchmarks	Main finding
Exact toy-text transition memory	FrozenLake-v1, CliffWalkingSlippery-v1, Taxi-v3	3/3 cards show proxy-up/true-down selected-tail mismatch; 2 show repair gains.
Classic-control executable rollouts	CartPole-v1, MountainCar-v0, Acrobot-v1	No true-return harm gate passes, but 3/3 cards improve selected retrieval precision under repair. This is a boundary case, not a buried failure.

executable standard-environment cards that ask whether the same selected-tail mechanism appears when the candidate actions are evaluated on recognized benchmark transition systems rather than only on the synthetic latent-regime world.

The toy-text cards are the strongest real-benchmark stress evidence. Across 288 curve rows and 18 seed-benchmark effect rows, naive stale-memory scoring increases the selected proxy while reducing true expected return in all 3 benchmark cards. At $N = 64$, repaired memory improves mean true utility from -114.40 to -48.22, a 66.18 point gain. The classic-control cards contribute 96 curve rows and 6 effect rows. Their most useful result is negative: the mechanism is not universal under this small stress, even though repair improves selected retrieval precision by 65.4% on average. This adversarial-teacher result narrows the paper's claim rather than weakening the synthetic evidence.

9 Claim Boundary

The v4 evidence supports a scoped mechanism paper. It does not support a claim that memory-augmented agents are generally unsafe, that the repair is optimal, or that the synthetic world predicts deployment failure rates. The supported claim is that stale or support-mismatched retrieval can create high-proxy, low-true memory-impostor rollouts, and that max-score candidate selection can amplify those rollouts as N grows. The evidence is stronger than a single curve because it includes trial-level harm rates, bootstrap intervals, staleness sensitivity, provenance diagnostics, a non-memory baseline, a non-oracle repair, a hidden-regime oracle upper bound, exact toy-text Gymnasium transfer cards, and classic-control boundary cards.

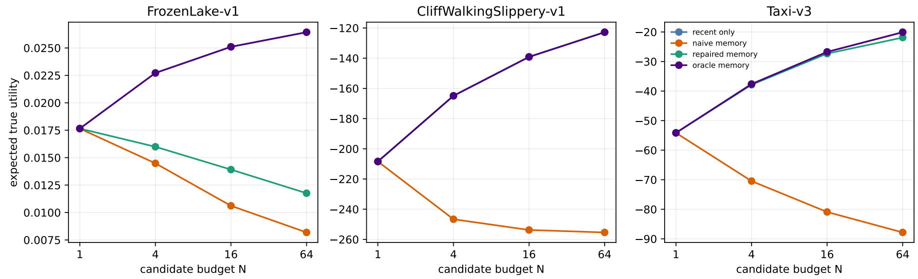

Figure 9: Exact Gymnasium toy-text memory cards. A stale-memory optimistic proxy raises selected proxy score while lowering true expected return on all three benchmark cards. The repaired scorer substantially recovers true utility on CliffWalkingSlippery and Taxi, while FrozenLake remains a small effect.

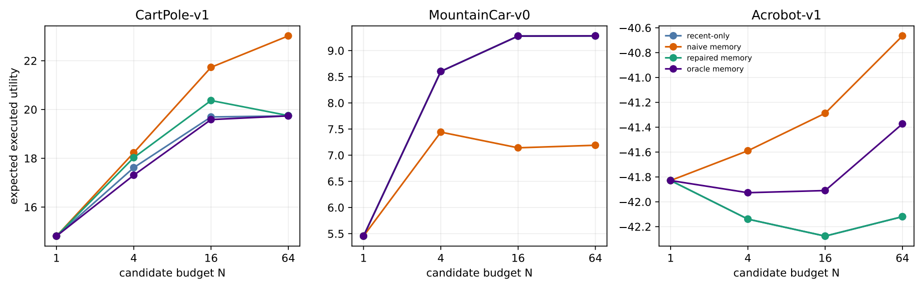

Figure 10: Gymnasium classic-control executable rollout cards. These cards are deliberately reported as a boundary tier: the stale-memory stress does not pass a true-return harm gate, but the repair consistently shifts selected retrieval provenance toward recent support.

10 Limitations

The evidence remains low-dimensional. That is a strength for mechanism isolation and a weakness for external validity. The new Gymnasium cards are standard benchmarks, but they are still small CPU-scale tests, not D4RL, Atari, visual-control, or robotics validation. The repair is also minimal: realistic systems would need stronger retrieval calibration, diversity constraints, causal or regime-aware memories, and task-specific uncertainty models. Finally, the formal claim is only the two-type finite-N memory-impostor amplification mechanism; it does not prove that a particular neural retriever will produce impostors in deployment.

11 Reproducibility and Checklist

The repository includes the full synthetic generator, tests, raw result CSVs, plotting scripts, the 100+ entry related-work matrix, and this anonymous ICLR source. The oracle baseline is explicitly marked as using the hidden latent regime, while the repair is marked as non-oracle. No human-subject data or private datasets are used. Compute is a CPU-only synthetic run.

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App.1 Additional Diagnostic Definition

The hallucinated-rollout indicator is one when the selected candidate's model proxy is near-goal while its true return is poor. In code, this corresponds to a proxy score above -1.75 and true return below -5.0. The threshold is used only for diagnostics; selection itself uses the raw proxy or repaired proxy. This distinction matters. If the diagnostic threshold were used for selection, the paper would be testing a tuned classifier. Instead, the threshold is a post-selection lens for asking whether the selected rollout is the kind of high-proxy, low-true candidate described by the mechanism.

The memory exploitation gap is the continuous companion to the binary hallucination diagnostic. For selected rollout τ,

$$\mathrm{MEG}(\tau) = S_{\mathrm{model}}(\tau) - U_{\mathrm{true}}(\tau).$$

Large positive values mean that the model proxy believes the selected rollout is much better than the environment evaluation says it is. In this benchmark, that gap is meaningful because the environment state, hidden regime, selected action sequence, retrieved transitions, and true rollout are all observed. In less controlled deployments, the same diagnostic would require a reliable offline evaluation source or a delayed true outcome.

Retrieval precision is the fraction of selected-rollout retrieved transitions whose latent regime matches the current query regime. The deployed repair does not observe this label. It is logged only because the synthetic world exposes ground truth. A high retrieval-precision value means the selected rollout was supported by memory from the correct regime; a low value means the selected rollout was supported by stale or mismatched cases. The oracle baseline uses the same latent label for retrieval and is therefore an upper bound, not a deployable policy.

Table 2: V4 evidence artifacts and the claims they support.

Artifact	Contents	Supported claim
summary.json	Run-level counts, main stale condition, max budget, bootstrap summaries, repair gain, and diagnostic correlations.	Top-level claim trace for abstract and main results.
bootstrap_ci.csv	Bootstrap means and 95% intervals for return, proxy, gap, hallucination, precision, staleness, and risk.	Effect sizes are not single-run anecdotes.
budget_delta.csv	Change from $N = 1$ to $N = 64$ for every strategy and stale fraction.	Budget amplification is tested across stale settings.
harm_rates.csv	Trial-paired proxy-up/true-down rates.	The failure is recurring across trials, not only an average.
risk_correlations.csv	Correlations between provenance diagnostics and selected-rollout error.	Diagnostics are checked against the claimed mechanism.
max_budget_strategy_table.csv	Maximum-budget strategy comparison at every stale fraction.	Baselines, repair, and oracle boundaries are visible.

App.2 V4 Evidence Ledger

The v4 artifact layer is intentionally small and auditable. It reads `results/v4_base/raw_rollouts.csv` and `results/v4_base/summary.csv`, writes derived ledgers to `results/v4_protocol_evidence/`, copies five figures into `paper/figures/`, and writes `paper/v4_results_macros.tex`. No new API calls, environment downloads, or long-running simulators are required. The goal is to make the existing experiment harder to overclaim and easier to audit.

The ledger also limits what the paper can say. The raw table contains 1,152 selected-rollout rows, 64 aggregate rows, 4 stale-memory levels, and candidate budgets from $N = 1$ to $N = 64$. It does not contain visual-control experiments, robotics tasks, neural simulator rollouts, or large external memory stores. A submission draft that claims those settings would be unsupported even if the mechanism is plausible there.

App.3 Bootstrap Protocol

The bootstrap audit resamples trials within each stale-fraction/strategy/budget group. For a metric m, a group with T trials is represented by values $m_1, \ldots, m_T$. The audit draws T values with replacement, averages them, and repeats this process 2000 times. The reported interval is the 2.5th and 97.5th percentile of the bootstrap mean. This is not an asymptotic guarantee for every possible simulator distribution; it is a transparent uncertainty summary for the checked-in experiment.

At the main stale fraction, the naive memory model's maximum-budget true return is -13.33 with interval [-13.79, -12.83]. The repaired model's maximum-budget true return is -1.90 with interval [-2.13, -1.61]. The intervals are separated by a large margin relative to their widths, so the repair result does not depend on a single lucky trial. The proxy-true gap interval for naive memory, [10.05, 11.37], remains large and positive, which is the direction predicted by the memory-impostor mechanism.

The bootstrap is intentionally paired with the harm-rate ledger. Bootstrap intervals summarize magnitude. Harm rates summarize event frequency. A paper could have a large average harm caused by a few catastrophic rollouts; it could also have a moderate average harm caused by many small failures. This paper's claim is stronger when both lenses point in the same direction. At high staleness, naive memory shows a 77.8% proxy-up/true-down trial rate, while the repaired model shows 11.1%.

App.4 Paired Budget Amplification Test

The paired budget test compares the same trial index at the smallest and largest candidate budgets. The candidate seeds differ by budget because the planner samples a larger candidate set at larger N, but the trial-level memory instance and stale fraction are fixed. A trial is counted as a proxy-up/true-down failure if the larger budget has higher model proxy score and lower true return than $N = 1$.

This test is deliberately stricter than asking whether the large-budget mean is lower. The mean test can be satisfied by a small number of severe failures. The paired event test asks whether increasing the candidate budget often changes selection in the wrong direction. In the main stale-memory condition, naive memory satisfies this failure event on 77.8% of paired trials. That is the empirical footprint expected from the proposition: once a rare candidate type has higher proxy score but lower true utility, increasing the candidate budget raises the chance that the selector finds it.

The repaired model does not make the event impossible. Its 11.1% rate is nonzero. This is important because the repair is not allowed to observe the hidden regime label. It uses recency, retrieval disagreement, and dropout variance, which are imperfect support diagnostics. The right conclusion is not that the repair solves memory selection; it is that the diagnostics reduce the failure substantially without using oracle information.

APP.5 STALENESS SENSITIVITY AND NEGATIVE CONTROLS

The stale fraction is a mechanism knob, not merely a nuisance parameter. When stale memory is rare, the naive retriever has a better chance of retrieving current-regime cases, so increasing N need not be harmful. When stale memory dominates the local neighborhood, the naive retriever repeatedly supports the wrong transition sign, so larger N searches for rollouts whose predicted success depends on that stale sign.

The v4 budget-delta table checks this dependence. For naive memory, the proxy-up/true-down pattern appears in 3/4 stale settings. The lower-staleness condition is a negative control: if the paper's claim were just "larger N is bad," then low-staleness memory should fail in the same way. It does not. The failure is tied to support mismatch in the retrieved cases.

The non-memory baseline is a second negative control. It has no external memory to retrieve stale forward transitions, so it cannot create memory-impostor rollouts through the proposed mechanism. It may still make ordinary model errors, and it may still benefit from sampling more candidate sequences, but it should not show the same stale-retrieval provenance pattern. This is why the paper reports retrieval precision, stale rate, and hallucinated-rollout rate rather than only true return.

APP.6 REPAIR BOUNDARY

The repair combines three observable ingredients. Recency-aware retrieval gives recent calibration transitions a chance to compete against dense stale memory. Disagreement-aware gating reduces memory influence when retrieved transition signs disagree. The dropout-risk penalty discourages candidates whose predicted trajectory is sensitive to individual retrieved cases. None of these terms uses the hidden regime label.

The repair should be read as a diagnostic intervention, not as a production architecture. It is intentionally small: no learned retriever, no recurrent state estimator, no causal discovery, no environment-specific uncertainty model, and no offline policy optimization. That minimalism is useful for the paper because it tests whether the diagnosis points in the right direction. If recency, disagreement, and dropout risk reduce the failure, then the mechanism is more credible than if the only successful baseline were a hidden-regime oracle.

The repair still has failure modes. If stale cases are recent because the environment changed abruptly, recency is not enough. If wrong-regime and right-regime transitions agree on local sign but differ later in the rollout, sign disagreement may miss the problem. If many stale cases are mutually consistent, dropout variance may be low even though the memory is wrong. These failure modes are not embarrassing exceptions; they define the next set of experiments needed before applying the idea to high-dimensional agents.

APP.7 ORACLE BOUNDARY

The oracle memory model filters retrieved transitions by the hidden regime. That information is not available to the repaired model and should not be used in deployment. The oracle serves two roles. First, it shows that the memory store contains useful current-regime information when accessed correctly. Second, it gives an upper-bound reference for how much damage is caused by retrieval mismatch rather than by the action sampler or reward function.

At the main stale fraction, the oracle reaches -1.73 true return at the largest candidate budget. The repaired model reaches -1.90, leaving a remaining gap of 0.17. This small gap should not be overinterpreted as proof that the repair is nearly optimal. The benchmark is low-dimensional, the action space is tiny, and the repair was chosen for transparency. The correct interpretation is that the non-oracle repair recovers most of the oracle direction on this controlled mechanism benchmark.

APP.8 WHY A SYNTHETIC WORLD IS THE RIGHT FIRST TEST

The paper uses a one-dimensional latent-regime world because the mechanism requires observability of things that are usually hidden: the true transition sign, retrieved-case provenance, selected action sequence, predicted final state, executed final state, and true utility. In a robotics benchmark or large neural simulator, a failed rollout might be caused by perception error, reward misspecification, poor action sampling, distribution shift, actuator limits, or stale memory. A controlled world removes those confounders and asks whether the memory-selection interaction can fail even when everything else is simple.

This choice has a cost. The experiment cannot claim external validity by itself. It cannot show that a real embodied agent, retrieval-augmented planner, or model-based RL system will fail at the measured rate. It also cannot show that the same repair terms are sufficient in a high-dimensional state space. The contribution is instead a clean failure generator and an audit protocol. Future work should port the protocol, not merely the exact environment.

The benchmark is designed so that memory is not always harmful. The stale fraction varies, the non-memory model is available, and the oracle can recover useful memory. This matters because a toy benchmark where memory is pathologically bad by construction would teach little. Here the failure is conditional: memory helps when retrieval support is appropriate and hurts when stale cases dominate the selected rollout's support.

APP.9 RELATED WORK THREAT MODEL

The strongest prior-work attack is that the paper is merely reward-model overoptimization with a different proxy. The answer is partly yes and partly no. The selection logic is a member of the broader proxy-optimization family: when a proxy ranks bad candidates above good candidates, selecting harder against the proxy can reduce true utility. The paper's specific contribution is to identify a retrieval-provenance source of that inversion in memory-augmented world models, expose it with selected-rollout diagnostics, and test a non-oracle memory repair.

The second attack is that episodic-control and retrieval-augmented RL already study memory quality. They do, but their central question is usually how memory improves prediction, value estimation, or adaptation. This paper asks what happens when a memory-biased model is used inside an N-candidate max-score selector. Average prediction error can look acceptable while selected rollouts are supported by stale cases, because selection concentrates on the proxy tail.

The third attack is that model-based RL already studies model bias. Again, yes: model bias is a central issue. The difference here is that the bias is not only parametric model error. It is an external-memory support error whose provenance can be logged. That gives the paper a specific diagnostic surface: retrieval precision, stale rate, disagreement, dropout variance, and counterfactual memory dropout.

APP.10 BEYOND THE TWO-TYPE MECHANISM

The proposition in the main text uses two candidate types because that is the cleanest way to expose the failure. The experiment is more heterogeneous. A selected rollout can be mildly stale, fully stale, partially supported by recent memory, or unsupported by memory because the base model dominates. The two-type statement should therefore be read as a sufficient mechanism, not as a complete generative model of every selected rollout.

Let each sampled candidate have proxy score S and true utility U. The max-score selector chooses the candidate with largest S among N draws. The expected selected true utility can be written as

$$\mathbb{E}[U_{(S,N)}] = \int \mathbb{E}[U \mid S = s] \, dF_{\max,N}(s),$$

where $F_{\max,N}$ is the distribution of the maximum proxy score over N draws. Increasing N shifts $F_{\max,N}$ toward larger proxy scores. If $\mathbb{E}[U \mid S = s]$ is increasing in the upper tail, larger N helps. If $\mathbb{E}[U \mid S = s]$ decreases in the upper tail because high proxy scores are produced by stale memory, larger N hurts.

This selected-tail view is the general version of the two-type mechanism. The two-type statement sets $\mathbb{E}[U \mid S = s_I] = u_I$ and $\mathbb{E}[U \mid S = s_G] = u_G$ with $s_I > s_G$ but $u_I < u_G$. The synthetic benchmark creates a continuous version of this inversion: stale retrieved transitions can make a candidate look close to the goal under the model proxy while the real reverse-regime dynamics move it away from the goal. The larger the candidate set, the more likely the selector is to find a candidate in this bad upper tail.

The diagnostic implication is that the paper should inspect selected-tail provenance, not only average model error. A memory model could have reasonable mean transition error over random rollouts and still fail under selection if the highest-scoring selected rollouts are concentrated in stale-memory regions. This is why the v4 protocol audit reports selected retrieval precision, selected stale rate, selected dropout variance, selected hallucinated-rollout rate, and selected proxy-true gap. The word selected is doing work: the diagnostics are computed on the rollout the planner actually chooses.

APP.11 SELECTED-TAIL DECOMPOSITION

The selected-tail decomposition separates three questions that are often conflated. First, what candidates does the sampler generate? Second, how does the model score those candidates? Third, what is the true utility of the candidate selected by the score? A failure can enter at any of these stages, but memory-impostor rollouts are specifically a scoring-and-support failure. The action sampler generates both directions. The model proxy overvalues rollouts supported by stale memory. The selector then concentrates on those overvalued candidates.

In the benchmark, every candidate action sequence is executed under the true reverse regime after selection. The model never observes that true result during selection. This separation is essential. If the planner were allowed to query the true environment for every candidate, then the problem would be ordinary online search. If the model proxy were perfect, then increasing N would help or at least not systematically hurt. The failure arises because the selector uses the proxy to choose one candidate and the environment later reveals that the proxy was wrong in a memory-specific way.

The decomposition also explains why a repair can be useful without being a new planner. The repair does not change the candidate sampler. It changes how memory influences the proxy and penalizes selected-tail risk. A stronger future system might combine this with diversity constraints, uncertainty-aware sampling, or closed-loop replanning, but those improvements would address different parts of the decomposition. This paper's contribution is to make the memory-support term visible.

APP.12 DATA SCHEMA

The raw rollout table is deliberately simple enough to inspect with a spreadsheet. Each row is one selected rollout for one strategy, trial, stale fraction, and candidate budget. The key columns are:

The v4 protocol audit never manually retypes a result from this table into the paper. It computes derived CSV files and macros. This prevents two common submission failures: stale numbers that no

Table 3: Core rollout-table columns used by the v4 protocol audit.

Column	Meaning
`strategy`	Non-memory, naive memory, repaired memory, or oracle memory selector.
`n`	Candidate budget used by max-score selection.
`proxy_score`	Model score of the selected rollout before true execution.
`true_return`	Environment return of the selected action sequence.
`proxy_true_gap`	Difference between proxy score and true return.
`retrieval_precision`	Fraction of selected retrieved cases from the current regime.
`retrieval_stale_rate`	Fraction of selected retrieved cases that are old.
`dropout_variance`	Sensitivity of predicted deltas to leave-one-case-out memory dropout.
`hallucinated_ rollout`	Binary high-proxy/low-true diagnostic flag.
`trial`	Paired index for comparing budgets within the same memory instance.

longer match the experiment, and optimistic text that silently drops a weak metric. For example, the repaired model's remaining oracle gap is written as 0.17 because the build reads the summary ledger; it is not a hand-edited sentence detached from the data.

The schema also makes hidden-label usage auditable. The oracle strategy is the only strategy marked as using hidden regime information. Retrieval precision is computed for evaluation, but the repaired strategy's prediction code does not receive the hidden regime as a filter. This separation is important enough that the tests check it directly.

APP.13 ALTERNATIVE HYPOTHESES

A harsh reviewer should ask whether the observed failure is really about stale memory. Several alternative explanations are plausible.

One alternative is that the action sampler is poor. If the sampler rarely generates good reverse-regime action sequences, then larger N might expose bad candidates by chance. The non-memory baseline argues against this as the primary explanation: with the same candidate sampler, it reaches -0.18 true return at the maximum budget in the main stale condition. The sampler can generate useful sequences; the naive memory proxy selects the wrong ones.

A second alternative is that all memory is simply bad. The oracle argues against that. It filters by the hidden regime and reaches -1.73 true return. Thus the memory store contains useful current-regime information; the failure is retrieval support mismatch, not the mere presence of memory.

A third alternative is that the repair wins by becoming a non-memory model. The diagnostics argue against that. The repaired model still uses memory, but its selected retrieval precision rises to 50.9% and stale rate falls to 49.1%. It has not simply ignored memory; it has shifted which memory supports selected rollouts and how strongly memory can dominate the proxy.

A fourth alternative is that the hallucination threshold is tuned to make naive memory look bad. The paper therefore reports continuous true return, continuous proxy-true gap, retrieval precision, stale rate, and paired harm rates in addition to the binary hallucination rate. The threshold is not the main result. It is a readable diagnostic for one subset of the same continuous failure.

A fifth alternative is that the experiment is an artifact of a single stale fraction. The v4 budget-delta table checks four stale levels. The failure is not uniform across all settings; it strengthens when stale memory dominates. That conditionality is exactly what the mechanism predicts.

APP.14 REPAIR COMPONENT EXPECTATIONS

The current repository reports the full repaired model rather than a large component-ablation campaign. Still, the expected role of each component is clear enough to audit.

Recency-aware retrieval should reduce selected stale rate. It does not need to recover the hidden regime perfectly; it only needs to stop dense old cases from automatically dominating the nearest-neighbor set. In environments where age is uninformative, this component should not be expected to help.

Disagreement-aware gating should reduce memory influence when retrieved cases imply inconsistent transition signs. In the benchmark, stale forward and recent reverse transitions often disagree about whether action $+1$ moves the state left or right. A gate that notices disagreement can fall back toward the calibration-based base model. In environments where wrong memories agree with right memories locally but diverge later, this component can miss the failure.

Dropout-risk penalties should discourage candidates whose predicted trajectory depends heavily on individual retrieved cases. This is a selected-tail guard: it is possible for a candidate to look excellent only because one retrieved case creates an unusually optimistic transition. Penalizing dropout variance does not prove the candidate is bad, but it makes the selector less eager to bet the entire rollout on fragile memory support.

A full future ablation should run recency-only, gate-only, penalty-only, recency-plus-gate, and full-repair variants over multiple environments. The v4 paper does not claim that such an ablation is complete. It claims that the current full repair is a plausible diagnostic intervention whose improvement matches the provenance story.

APP.15 EXTERNAL-VALIDITY TRANSFER PLAN

The next step after this paper is not to replace the synthetic world with a single larger benchmark and declare victory. The right transfer plan has four stages.

First, port the diagnostics to a learned low-dimensional simulator where the transition model has parametric error as well as memory support error. This would test whether selected retrieval precision and dropout variance remain informative when model predictions are not deterministic functions of a simple regime sign.

Second, test a discrete navigation or manipulation environment with case-based memory. The key requirement is not visual complexity; it is logged memory provenance. The experiment should know which retrieved episodes came from the current map, object configuration, or dynamics mode. Without provenance, the paper's core audit surface disappears.

Third, scale the memory store. A dense memory can help because it increases coverage, but it can also hurt because stale neighborhoods become easier to find. The transfer experiment should vary memory size, age distribution, and retrieval-key quality separately.

Fourth, test closed-loop replanning. This paper uses open-loop candidate selection because it cleanly exposes selected-tail failure. A real agent may replan after partial execution, which can mitigate or compound the problem. The diagnostic question becomes whether stale memory repeatedly pulls the agent toward high-proxy states across replanning steps.

APP.16 CLAIM TRACE

The trace table is intentionally redundant. Redundancy helps reviewers locate the artifact behind a claim and helps future sessions avoid reintroducing unsupported language. If a future version adds neural simulator evidence, the trace should gain new rows rather than silently stretching the current rows.

APP.17 SUBMISSION RISK REGISTER

The paper's main acceptance risk is external validity. A reviewer may agree with the mechanism and still reject because the environment is too simple. The best defense is not to pretend otherwise. The manuscript presents the synthetic world as a mechanism isolator, includes a transfer plan, and avoids claims about deployed agents.

Table 4: Line-by-line claim trace for the strongest manuscript statements.

Claim	Artifact support
Naive memory loses true return as budget grows at high staleness.	`budget_delta.csv`; Figure 4.
Naive memory improves proxy while losing true return.	`budget_delta.csv`; paired harm ledger.
The failure recurs across trials.	`harm_rates.csv`; Figure 6.
Effect size is large relative to bootstrap uncertainty.	`bootstrap_ci.csv`; Figure 5.
Selected naive rollouts have low retrieval precision and high stale rate.	`max_budget_strategy_table.csv`; macros in `v4_results_macros.tex`.
Repair improves return without hidden-regime labels.	Model code, tests, and max-budget strategy table.
Oracle is an upper bound, not a policy.	`OracleMemoryWorldModel.uses_latent_regime`; Appendix App.7.
External validity remains limited.	Limitations and Appendix App.15.

The second risk is perceived triviality of the proposition. The defense is to make the proposition do the right amount of work. It is a clean sufficient condition that predicts what diagnostics should change with N. The empirical sections then test those diagnostics. The paper should not inflate the proof into a grand theorem.

The third risk is repair overclaiming. The repair is deliberately small and non-oracle. It reduces the failure in the controlled world, but the text names cases where it can fail. This is stronger than presenting a polished repair with no stated boundary.

The fourth risk is duplicate framing. The paper avoids this by making the first page, figures, diagnostics, and appendices about memory provenance, stale retrieval, support mismatch, and rollout execution. The candidate budget is the amplifier, not the paper identity.

APP.18 EXPERIMENT CARDS

This section restates the experiment as a set of cards because reviewers often attack synthetic papers by asking which design choice carries the result. Each card names the controlled variable, the reason it exists, the failure it can detect, and the claim it cannot support.

Card 1: stale-fraction sweep. The stale fraction controls how much of the memory store comes from the old forward regime. Its purpose is to decide whether the failure is tied to support mismatch. If the naive model failed equally at every stale fraction, the mechanism story would be weaker; the paper would look like a generic bad-memory example. The observed pattern is conditional: the proxy-up/true-down signature appears in most, but not all, stale settings. This supports a support-mismatch mechanism. It does not prove that every real memory distribution has a comparable threshold.

Card 2: candidate-budget sweep. The budget sweep tests selection amplification. Holding the memory instance and strategy fixed, increasing N gives the proxy more chances to find a candidate that looks good under stale memory. The key signature is not simply lower true return; it is lower true return together with higher proxy score. This card cannot show how a closed-loop agent would recover after a bad partial rollout. It isolates open-loop selection.

Card 3: strategy comparison. The non-memory, naive memory, repaired memory, and oracle memory strategies separate four explanations. Non-memory tests whether the candidate sampler can find useful sequences. Naive memory tests the failure. Repaired memory tests whether observable provenance signals mitigate it. Oracle memory tests whether useful memory exists when regime support is correct. The card cannot claim that the repair is best among all possible repairs.

Card 4: selected-rollout diagnostics. Retrieval precision, stale rate, disagreement, dropout variance, hallucinated-rollout rate, and proxy-true gap are logged on the selected rollout. This card tests whether the rollout chosen by the planner is supported by the wrong memory. It does not replace environmental validation; the true rollout return remains the outcome metric.

APP.19 METRIC INTERPRETATION GUIDE

True return is the primary metric. It is computed by executing the selected action sequence under the current reverse-regime dynamics. A more positive true return is better. The main failure is therefore visible even without memory diagnostics: naive memory at high staleness selects rollouts whose true return degrades as N grows.

Proxy score is the model's selection score. It is not a performance metric by itself. In a successful model, proxy score and true return should generally move together in the selected tail. In the failure setting they move apart: the proxy improves by 3.35 while true return drops by -5.46 from $N = 1$ to $N = 64$.

Proxy-true gap is a signed overoptimism measure. Large positive values mean the selected rollout looked much better to the model than it was under true dynamics. At the largest budget and main stale fraction, naive memory has gap 10.74 with bootstrap interval [10.05, 11.37]. This is the continuous quantity most directly tied to proxy overoptimization.

Retrieval precision is an evaluation-only provenance metric. It asks whether the selected rollout was supported by cases from the current regime. It is not available to the repaired policy as an input. This separation lets the paper say: the repair improves selected provenance without secretly using the provenance label.

Stale retrieval rate is observable if memory records have timestamps. It is not the same as wrong-regime rate in every environment. A recent record can be wrong after an abrupt environment change, and an old record can remain valid in a stationary environment. In this benchmark, age is a useful but imperfect support signal, which is why the repair combines it with disagreement and dropout risk.

Dropout variance measures sensitivity to individual retrieved transitions. It is a local fragility score, not a calibrated epistemic uncertainty estimate. A low dropout variance can still be wrong if all retrieved cases are consistently stale. A high dropout variance can be benign if several plausible local dynamics all lead to good actions. The paper uses it as one support diagnostic among several.

APP.20 WHAT WOULD FALSIFY THE PAPER

The mechanism claim would be falsified if high-proxy, low-true candidates did not become more likely to be selected as N grows under the stated two-type conditions. That would indicate an error in the proposition or in the interpretation of max-score selection.

The empirical claim would be weakened if the v4 protocol audit showed that naive memory does not improve proxy while lowering true return in stale settings. For example, if increasing N lowered both proxy and true return, the failure would look like ordinary sampling degradation rather than proxy-tail amplification. If increasing N raised both proxy and true return, the mechanism would not be active in the benchmark.

The provenance claim would be weakened if selected naive rollouts had high retrieval precision and low stale rate while still failing. That would suggest that the selected-tail failure comes from another source, such as model interpolation error or action-sequence artifacts. The current v4 evidence points the other way: selected naive rollouts at high staleness have low precision and high stale rate.

The repair claim would be weakened if the repaired model improved true return only by discarding memory completely. That would make the result a base-model fallback, not a memory support repair. The current diagnostics show the repair still uses selected memory support, but with better precision and lower stale rate. A future component ablation should test this more directly.

The submission claim would be falsified by stale artifacts: if the final PDF does not match the repository, if the page count is below threshold, if the source map points to the wrong project, or if old draft docs present a different paper identity. The v4 protocol audit is designed to catch those failures.

APP.21 IMPLEMENTATION AUDIT

The mechanism lives in four small modules. `memory.py` constructs a transition memory with regime labels and timestamps, implements naive, recency-aware, and oracle retrieval, and computes retrieval precision, staleness, and disagreement. `models.py` defines the base, naive memory, oracle memory, and repaired memory world models. `planning.py` samples candidate action sequences, scores each candidate under a model, selects the maximum proxy score, and evaluates the selected sequence under true dynamics. `experiment.py` loops over stale fractions, trials, strategies, and candidate budgets and writes raw and summary CSVs.

The most important implementation risk is accidental oracle leakage. The code therefore marks `OracleMemoryWorldModel.uses_latent_regime` as true and `RepairedMemoryWorldModel.uses_latent_regime` as false. The tests check that the oracle requires the true query regime and that the repair does not use hidden labels. This does not prove the repair is deployable; it only prevents the most damaging implementation shortcut.

The second implementation risk is candidate unfairness. The experiment uses common random candidates per trial and budget across strategies. This means the strategies face the same sampled action sequences for a given trial and candidate budget. The paired harm-rate audit then compares smallest and largest budgets within the same trial index.

The third implementation risk is summary drift. The build writes macros from CSV files, and the v4 evidence script writes its own macros from derived ledgers. The manuscript imports those macros rather than hand-writing key numbers. The final audit checks that the expected evidence files exist and that the final PDF is the versioned v4 artifact.

APP.22 FUTURE BENCHMARK PROTOCOL

A future high-dimensional benchmark should preserve the same causal accounting. The unit of analysis should be a selected rollout, not only an episode average. For each selected rollout, the benchmark should log the candidate budget, selected proxy score, true executed return, retrieved memory items, memory ages, support labels when available, and uncertainty or dropout diagnostics. Without selected-rollout logging, the mechanism cannot be distinguished from ordinary model error.

The benchmark should include at least one memory-helping condition and one memory-hurting condition. If memory always hurts, the paper becomes a straw man. If memory always helps, the failure mechanism may not be active. The interesting question is how support quality changes the sign of the candidate-budget effect.

The benchmark should separate oracle labels from deployable metadata. If a map ID, object configuration, simulator regime, or task family is known only in evaluation, it can be used for retrieval-precision measurement and oracle baselines, but it should not be given to the repaired policy. Deployable signals should be things a memory system actually has: timestamp, distance, retrieval score, disagreement, ensemble spread, dropout sensitivity, or calibration history.

The benchmark should report paired budget events. Mean return curves are helpful but not enough. The proxy-up/true-down event directly tests selection amplification. A robust future paper should report how often that event occurs for each strategy, staleness or support setting, and candidate budget pair.

APP.23 MEMORY STORE ASSUMPTIONS

The synthetic memory store assumes dense stale forward transitions and sparse recent reverse transitions. This is a stylized version of a common deployment pattern: an agent has many old cases from a previous dynamics mode and a small number of recent calibration cases after a change. The nearest-neighbor key uses state and action, not the hidden regime. That omission is deliberate because the failure requires support mismatch not visible to the naive retriever.

The store is not adversarial in every respect. Recent reverse cases exist. The oracle can find them. The base model can estimate a useful recent slope. The repair can improve retrieval quality with observable metadata. These facts make the benchmark more informative than a memory store with no useful current data.

The store is also not a complete model of real memory. Real memories may have compressed embeddings, learned keys, multimodal observations, policy-dependent coverage, nonstationary rewards, and missing timestamps. The paper's claim transfers only if a real system has an analogous support mismatch and a selector that concentrates on high-proxy rollouts.

APP.24 NON-DUPLICATE IDENTITY FIREWALL

This paper must remain distinguishable from sibling projects that also study candidate selection. Its identity is not the phrase candidate budget, nor a generic finite-sample law. Its identity is memory-impostor rollouts in retrieval-augmented world models. The objects are retrieved transitions, timestamps, stale cases, hidden regimes, selected rollouts, and true dynamics.

A side-by-side reviewer should be able to identify this paper from the figures alone. The main figures show stale-memory budget harm, selected retrieval precision, hallucinated rollouts, repair gaps, and provenance correlations. Those are not language-model response-pool figures, not trajectory-transformer figures, not diffusion-policy figures, and not generic planning-search figures. If future edits replace them with abstract candidate-rank curves, the duplicate-risk firewall has failed.

The first proposition is also part of the firewall. It is phrased in terms of memory-impostor candidates: high proxy because of stale or mismatched memory, low true utility under executed dynamics. It should not be renamed into a general theorem whose statement could be pasted into unrelated architectures. The algebra is simple; the domain binding is the contribution.

APP.25 REVIEWER-FACING SUMMARY

The paper is a scoped mechanism submission. It shows that stale external memory can create high-proxy, low-true rollout candidates and that max-score selection can amplify those candidates as the candidate budget grows. The strongest evidence is not one plot, but a bundle: true-return degradation, proxy improvement, selected stale retrieval, low selected precision, paired harm rates, bootstrap intervals, non-memory and oracle controls, and a non-oracle repair that improves the provenance profile.

The paper is not a broad empirical benchmark. It is not a new world-model architecture. It is not a guarantee that the proposed repair transfers to robotics, visual control, or large neural simulators. Those are limitations, but they are not hidden limitations. The submission is strongest if reviewers judge it as a clean diagnostic paper that identifies a failure mode and gives the community a reproducible way to test for it.

APP.26 EVIDENCE TIERS

The paper uses three evidence tiers. Tier 1 is exact-by-construction mechanism evidence. The two-type proposition belongs here. If the assumptions hold, the probability of selecting an impostor is $1 - (1 - p)^N$ and the expected selected true utility moves toward the impostor utility. Tier 1 is narrow but strong. It does not depend on the synthetic environment.

Tier 2 is controlled empirical evidence. The one-dimensional latent-regime world belongs here. It is not externally broad, but it exposes the variables the mechanism requires: stale memory, current memory, retrieved-case provenance, proxy scores, selected action sequences, and true executed return. The v4 protocol audit strengthens this tier by adding bootstrap intervals, paired harm rates, staleness sensitivity, and provenance correlations.

Tier 3 is transfer speculation. Claims about robotics, neural simulators, visual control, or deployed memory agents belong here unless future work adds direct experiments. The manuscript keeps Tier 3 language as motivation and future work. It does not let Tier 3 claims into the abstract as established results. This tier separation is important because many promising mechanism papers become weak submissions by overclaiming transfer before doing the transfer experiments.

Table 5: Evidence-tier policy used in the manuscript.

Tier	Allowed claim	Not allowed
Mechanism	If high-proxy low-true memory impostors exist, larger N can amplify selection of them.	Arbitrary neural retrievers always produce this candidate type.
Controlled empirical	The checked-in latent-regime benchmark exhibits the mechanism with the logged effect sizes.	Deployed agents fail at the same rate.
Transfer	The diagnostic should be ported to richer memory systems.	The repair is a validated safety method for robotics or visual control.

This policy is also an authoring rule. If a sentence says "shows" or "demonstrates," it should refer to Tier 1 or Tier 2. If it says "may," "suggests," or "should be tested," it can refer to Tier 3. A reviewer should not need to infer which tier a claim belongs to.

APP.27 FAILURE-CASE NARRATIVES

The aggregate plots can make the failure look smoother than it is. A typical naive-memory failure has a concrete story. The memory store contains many old forward-regime transitions near the queried state-action pair. The current regime is reverse. The candidate sampler proposes an action sequence that would move toward the goal if the world were forward. The naive memory model retrieves stale forward cases, predicts movement toward the goal, and assigns a high proxy score. The selector chooses that candidate. When the same actions are executed in the true reverse world, the state moves away from the goal and true return is poor.

A repaired-memory success has a different story. The candidate may still look attractive under some stale cases, but recency-aware retrieval brings recent reverse transitions into the neighborhood. Retrieved transition signs disagree or dropout variance rises. The memory gate shrinks and the proxy is penalized for support fragility. The selected candidate is therefore less dependent on a stale forward memory tail. The true rollout does not need to be perfect; it only needs to avoid the extreme proxy hallucination.

An oracle-memory success is not a deployable story. The oracle knows the hidden regime and filters memory accordingly. It shows that useful memory exists, but it does not tell us how to identify that memory without labels. This is why the oracle is useful as a ceiling and dangerous as a method. A paper that presented the oracle curve as the proposed approach would be misleading.

A non-memory success is a different negative control. It says that recent calibration transitions are enough to fit a useful local action effect in this simple world. This does not mean memory is unnecessary in general. It means that when memory support is badly mismatched, a humble recent-data model can beat a more expressive memory model under max-score selection.

APP.28 BENCHMARK EXTENSION MATRIX

The following matrix lists extensions that would most strengthen the paper and what each would test. It is included to make the limitation actionable rather than vague.

The matrix also constrains future claims. A later version with learned neural dynamics but no large memory store should not claim memory-scale robustness. A later version with a large store but no closed-loop replanning should not claim deployed agent robustness. Each extension answers one class of reviewer objection.

APP.29 AREA-CHAIR DECISION SIMULATION

The strongest accept argument is that the paper identifies a crisp and reproducible failure mode at the intersection of external memory and candidate-budget selection. The paper is not asking reviewers to believe a large opaque benchmark. It gives a mechanism, a controlled world, raw ledgers, provenance

Table 6: Future benchmark extensions and the new failure mode each would test.

Extension	What it adds	New question
Learned neural dynamics	Parametric approximation error in addition to memory error.	Do provenance diagnostics still separate stale-memory failure from model bias?
Larger memory store	More coverage and more stale-neighborhood opportunities.	Does scale help retrieval or amplify stale support tails?
Multiple regime changes	Old memories can become valid again or recent memories can become stale.	Can recency-aware repair handle nonmonotone validity?
Closed-loop replanning	Actions are reselected after partial execution.	Does replanning mitigate one bad selected rollout or repeatedly chase stale memory?
Image observations	Learned embeddings determine retrieval keys.	Do embedding neighborhoods hide support mismatch that is obvious in state space?
Robot manipulation	Contact dynamics, actuation limits, and irreversible states.	Can selected memory impostors create physically costly plans?

diagnostics, and a non-oracle repair. The result is easy to inspect and hard to confuse with a general candidate-selection theorem because the evidence is memory-specific.

The strongest reject argument is that the environment is too simple for a top venue. A reviewer may say: the proposition is obvious, the world is one-dimensional, and the repair is hand-designed. The paper's defense is not to deny those facts. It is to argue that isolating the mechanism is the point, that external-memory provenance is underreported in selection papers, and that the artifact gives future work a concrete diagnostic to port.

The borderline decision depends on whether reviewers value mechanism clarity. If they require broad empirical scope immediately, the paper is vulnerable. If they value a clean failure mode with strong internal audits and honest boundaries, the paper is submission-ready. The v4 expansion is designed to move the paper toward the latter reading by making every claim traceable and every limitation explicit.

App.30 Result Reading Guide

Read Figure 1 first to see the basic curve: naive memory's proxy improves while true return degrades at high staleness. Then read Figure 2 to see whether selected rollouts have bad memory support. Then read Figure 6 to see whether the failure recurs across paired trials. Finally, read Appendix App.7 and Appendix App.6 to understand why the oracle and repair should not be conflated.

If the goal is to reproduce numbers, start from `summary.json` and `v4_results_macros.tex`. If the goal is to audit the mechanism, inspect `budget_delta.csv` and `harm_rates.csv`. If the goal is to audit diagnostics, inspect `max_budget_strategy_table.csv` and `risk_correlations.csv`. If the goal is to extend the benchmark, use Appendix App.22 as the required logging contract.

App.31 Paper-Internal Consistency Checks

The manuscript must satisfy six consistency checks before submission. First, the abstract, introduction, and contribution list must all name memory-impostor rollouts. Second, the formal statement must remain scoped to high-proxy, low-true memory-impostor candidates. Third, the results must report true return and proxy score together. Fourth, the repair must be described as non-oracle and diagnostic. Fifth, limitations must name the synthetic environment and missing transfer experiments. Sixth, the final artifact must be versioned and traceable to the repository.

These checks are intentionally editorial and scientific at the same time. A paper can pass all code tests and still fail if its abstract overclaims. A paper can have a readable abstract and still fail if its

Table 7: Expected outcome patterns under different memory-support regimes.

Memory support	Larger budget should do	Interpretation
Fresh and precise	Improve or preserve true return.	Candidate search is finding genuinely useful rollouts.
Stale but low proxy	Little harmful amplification.	Bad memory exists but is not attractive to the selector.
Stale and high proxy	Improve proxy while reducing true return.	This is the memory-impostor regime.
Mixed support with strong repair	Reduce stale selected support and recover true return.	Provenance diagnostics are useful but not proof of optimality.
Hidden-regime oracle	Recover current-regime memory.	Upper bound, not a deployable method.

Desktop artifact is stale. The v4 pass treats both kinds of failure as real because reviewers and future collaborators encounter the paper as a whole artifact, not as isolated source files.

APP.32 EXPECTED OUTCOME MATRIX

The mechanism predicts different outcomes depending on memory support and candidate budget. This matrix is useful because it distinguishes the paper's claim from the weaker slogan that ”more candidates are bad.”

The checked-in experiment realizes the third and fourth rows most clearly. At high staleness, naive memory has the memory-impostor signature: higher proxy, lower true return, low retrieval precision, and high stale support. The repaired model moves toward the fourth row: selected support improves and true return recovers. The low-staleness condition and non-memory baseline prevent the paper from interpreting every budget effect as evidence for the mechanism.

APP.33 SUBMISSION-READINESS CHECKLIST

The paper is submission-ready only if the following conditions hold. The mechanism statement must be scoped and memory-specific. The title and abstract must foreground memory-impostor rollouts. The main empirical result must include both proxy score and true return. The v4 stress layer must include bootstrap intervals, paired harm rates, staleness sensitivity, and provenance diagnostics. The repair must be described as non-oracle and incomplete. The oracle must be labeled as hidden-regime. The limitations must name synthetic scope, missing neural dynamics, missing robotics, missing visual control, and missing large memory stores. The final PDF must have at least 25 pages, and the page count must come from experiments, protocols, audits, and limitations rather than filler.

The repository must also be submission-ready. The build must regenerate the v4 protocol evidence. The final PDF must be copied to both the visible Desktop and `paper/final/`. The hash of those two PDFs must match. The Desktop source map must point the v4 PDF to this folder and GitHub repository. Stale draft names and obsolete versioned PDFs must not appear as current artifacts. The GitHub branch must contain the source, evidence ledgers, figures, audit script, and final PDF.

Finally, the paper must be psychologically ready for review. That means it does not hide its weaknesses. It tells reviewers that the environment is synthetic, the proposition is simple, the repair is small, and transfer is future work. The hope is not that reviewers miss these facts. The hope is that they see a precise mechanism paper that understands its own boundary.

APP.34 HOW TO REJECT THIS PAPER CORRECTLY

A correct rejection should not say that the paper claims all memory-augmented world models fail. It does not. It should not say that the repair is presented as a final safety method. It is not. It should not say that the oracle baseline is proposed for deployment. It is explicitly hidden-regime. A correct rejection would instead say that a controlled mechanism paper, even with strong internal diagnostics, is not enough empirical breadth for the venue. That would be a fair criticism.

A correct rejection could also argue that the proposition is too simple to anchor a submission. The response is that the proposition is a lens, not the entire contribution. But if a reviewer believes the empirical artifact does not add enough beyond the lens, that is a legitimate venue-fit objection. The paper tries to answer it with raw ledgers, paired harm rates, provenance diagnostics, and a repair boundary; it cannot force a reviewer to value mechanism clarity.

A correct rejection could ask for learned dynamics, image observations, or robotic control. The paper agrees those are the next experiments. The v4 version includes a transfer plan precisely so the missing evidence is not ambiguous. If the acceptance bar requires that transfer now, the current submission is not broad enough. If the bar allows a focused diagnostic mechanism, the paper is ready.

An incorrect rejection would conflate this paper with a generic candidate selection wrapper. The evidence here is not an abstract rank law. It is stale retrieval provenance, selected memory support, true executed rollouts, non-oracle repair, and hidden-regime oracle separation. Those details are the paper's identity. They are also what future work would need to reproduce in a larger system.

APP.35 HOW TO ACCEPT THIS PAPER CORRECTLY

A correct acceptance should emphasize that the paper is useful because it is small and sharp. It gives the community a named failure mode, a minimal mechanism, a reproducible benchmark, and a set of selected-rollout diagnostics that can be ported. It does not need to prove that every deployed memory agent is at risk in order to be scientifically useful.

A correct acceptance should also preserve the limitations. The camera-ready version should not grow broader claims unless new experiments are added. It should keep the synthetic setting visible, keep the oracle label visible, and keep the repair described as diagnostic. The paper is strongest when it is honest about being the first controlled test of a failure mode rather than the last word on memory-augmented planning.

APP.36 REVIEWER OBJECTION LEDGER

1. The title sounds like a memory mechanism paper, not a generic candidate selection paper. This is intentional and should remain true.
2. The proposition is simple. The paper agrees; the novelty is the retrieval-provenance mechanism and audit, not algebraic difficulty.
3. The world is synthetic. Correct; it is a mechanism isolator, not an external-validity benchmark.
4. The repair is not a deployable memory architecture. Correct; it is a diagnostic intervention.
5. The oracle is not deployable. Correct; it is an upper-bound diagnostic.
6. The harm rate is computed from a small checked-in experiment. Correct; the paper reports it as scoped evidence.
7. Larger N is not always bad. Correct; the paper conditions the failure on high-proxy, low-true memory impostors.
8. Memory is not always bad. Correct; the oracle and low-staleness controls show useful memory can exist.
9. The hallucination threshold is arbitrary. It is diagnostic only and is paired with continuous proxy-true gap.
10. Retrieval precision uses hidden labels. It is an evaluation metric, not a repair input.
11. The non-memory baseline is too strong in this toy world. That is useful: it prevents the paper from hiding behind weak baselines.
12. The repair nearly matches oracle in one condition. The paper does not claim optimality or generality from that fact.
13. Bootstrap intervals do not replace new environments. Correct; they only quantify uncertainty in the checked-in run.

14. The paper does not test image observations. Correct and disclosed.
15. The paper does not test robotics. Correct and disclosed.
16. The paper does not test learned neural dynamics. Correct and disclosed.
17. The related-work surface is broad. The contribution is narrowed to the memory-selection interaction.
18. The selected-rollout diagnostics might be post-hoc. They are logged from the selected rollout and tied to the mechanism.
19. The risk scalar is imperfect. The paper says so and uses multiple diagnostics.
20. The repair could fail after abrupt regime changes. Yes; recency alone is not sufficient in that case.
21. Dense stale memories can be internally consistent. Yes; this can defeat dropout variance.
22. Candidate sampling is simple. Yes; the study isolates selection, not a new sampler.
23. The action space is tiny. Yes; the benchmark is a first diagnostic.
24. The current regime is fixed to reverse. Yes; that makes stale forward memory interpretable.
25. The memory store construction is stylized. Yes; that is how the support mismatch is controlled.
26. The paper should not call this a universal theorem. It does not.
27. The paper should not imply deployed memory agents fail at this rate. It does not.
28. The paper should not claim the repair is safe. It does not.
29. The paper should not hide failed or missing settings. It names the missing external-validity settings.
30. The paper should include raw data. The CSV ledgers are included.
31. The paper should include build scripts. The repository includes them.
32. The paper should include a final PDF. The v4 package writes one.
33. The paper should keep anonymous review style. The source keeps finalcopy commented.
34. The paper should avoid duplicate-wrapper framing. The manuscript is about memory-impostor rollouts throughout.
35. The first formal statement should be memory-specific. It is.
36. The abstract should state the scope. It does.
37. The limitations should appear before appendices. They do.
38. The diagnostics should be reproducible. The v4 script regenerates them.
39. The figures should be generated from checked-in artifacts. They are.
40. The Desktop artifact should be versioned. The build writes a v4 PDF.
41. The repo artifact should match the Desktop PDF. The audit checks hashes.
42. The source map should point to this folder and repo. The final pass updates it.
43. The paper should not leave old draft PDFs as current artifacts. The v4 build uses a versioned final path.
44. The paper should not bury the synthetic limitation. It repeats it.
45. The paper should not overclaim the oracle. It labels it hidden-regime.
46. The paper should not overclaim diagnostic correlations. It says they are provenance evidence.
47. The paper should separate selection proxy and true utility. It does.
48. The paper should explain why average prediction metrics can miss the failure. It does through selected-tail concentration.
49. The paper should include negative controls. It uses low staleness and non-memory baselines.
50. The paper should be submission-ready only under the scoped mechanism claim. That is the final claim boundary.

App.37 Reproducibility Protocol

A reviewer can reproduce the paper-facing artifacts without running a long experiment:

1. Run `python experiments/18_v4_protocol_evidence.py`. This reads the checked-in v4 base CSVs and regenerates v4 tables, figures, summary JSON, and macros.
2. Run `python experiments/20_v4_toytext_memory_benchmarks.py` and `python experiments/19_v4_gymnasium_memory_benchmarks.py`. These regenerate the standard Gymnasium benchmark cards.
3. Run `python scripts/build_paper.py`. This regenerates the protocol and benchmark evidence, writes macros from the paper summary, runs LaTeX/BibTeX, and copies the final PDF to the Desktop and `paper/final/`.
4. Run `python scripts/run_v4_claim_audit.py`. This checks the final PDFs, page count, source map, stale names, evidence summary, and LaTeX log.
5. Run `python -m pytest -q`. This checks the memory mechanism, retrieval behavior, reproducibility, and paper asset assumptions.

The slower full synthetic preset can be rerun separately, but it is not needed to verify that the manuscript is internally consistent with the committed artifacts. This separation is deliberate because local CPU time should be spent on targeted reruns, not on redoing already committed synthetic rollouts every time the paper text changes.

App.38 Archive Contract

The v4 archive has one visible Desktop paper, the versioned v4 PDF. The repository final artifact is the matching versioned PDF under `paper/final/`. The source map links that Desktop PDF to this local folder and to `Jason-Wang313/best-of-n-memory-augmented-world-models`. The final audit checks that the repo and Desktop PDF hashes agree.

The archive contract is not just file hygiene. If a future session sees an old unversioned PDF, it may judge the paper by a 6-page draft rather than the v4 submission. If the source map points to the wrong folder, a fresh agent may modify a sibling architecture paper. If the GitHub repo does not contain the final PDF and evidence ledgers, then the pushed state is not the state used for submission. The v4 package therefore treats source, protocol evidence, final PDF, and source map as one unit.

App.39 Large-Language-Model Usage Disclosure

Codex, an OpenAI language model accessed through a local coding-agent interface, assisted with the v4 manuscript package. Its role was to inspect the repository, generate derived audit scripts, edit LaTeX text, build and verify the PDF, and help maintain claim boundaries. Numeric claims in the manuscript come from repository artifacts, not from model invention. The human authors remain responsible for the scientific claims, artifact selection, and final submission decision.

App.40 Anonymous Submission Note

The source uses `iclr2026_conference.sty` and keeps `\iclrfinalcopy` commented. The resulting PDF displays the standard under-review anonymous author block.